

Intel Crescent Island

Xe3P data-centre inference GPU — consolidated specification, derived throughput model, memory architecture, model-fit analysis and software stack. Prepared as a companion reference to the Intel Xeon 6 Workload SKU workbook.

Version 1.0 · **Compiled** 9 August 2026 · **Product status** Pre-launch; customer sampling H2 2026, volume availability 2027 · **Scope** Single-card and 8-card node level; agentic / LLM inference serving
Critical caveat Intel has published *no* compute throughput figures for this part. All TFLOPS/TOPS values in §5 are architecturally derived scenarios, not specifications.

0 · Provenance discipline

Every quantitative claim in this document carries a provenance tag. This part is pre-launch and the public record contains a mix of Intel statements, board leaks and third-party arithmetic — conflating them would be misleading in a procurement context.

TAG	MEANING	CONFIDENCE	USE IN PLANNING
[I]	Intel-published (newsroom, Computex/OCP keynote, ARK)	High	Safe to plan against
[D]	Derived here from [I] data plus a stated architectural model	Medium	Use as a band, not a point value
[L]	Leak / unofficial (PCB photos, leaker claims)	Low-Medium	Directional only; re-verify at launch
[3P]	Third-party analysis or press arithmetic	Medium	Cross-check against [I] before use

READ THIS BEFORE QUOTING ANY NUMBER

Intel confirmed at Computex 2026 that it is **not** disclosing raw throughput specifications at this stage of development. Anyone presenting a Crescent Island TFLOPS figure today is deriving it. The §5 model is transparent about its assumptions so that the numbers can be corrected in one step once Intel publishes; do not propagate them into a bid or business case without the accompanying uncertainty band.

1 · Product identity and positioning

Crescent Island is Intel's re-entry into discrete data-centre AI silicon after the Gaudi line was wound down. The design thesis is deliberately narrow: rather than chase training FLOPS, it optimises for **memory capacity per dollar and per watt** in an air-cooled PCIe form factor, targeting steady-state inference and agentic serving where model residency and context length gate throughput more than peak matrix rate.

ATTRIBUTE	VALUE	SRC
Codename / class	Crescent Island — Intel Data Center GPU	[I]
Microarchitecture	Xe3P (performance-optimised derivative of Xe3; internally 'Celestial' lineage)	[I]
Primary workload	AI inference; explicitly positioned as "built for agentic AI"	[I]
Secondary workload	Scientific/numeric compute via FP64 datapath	[I]
Cooling class	Air-cooled enterprise servers (no liquid loop required)	[I]
Target buyer	Tokens-as-a-service providers; on-prem enterprise inference	[I]
Announced	14 Oct 2025, OCP Global Summit	[I]
Detail update	Computex 2026 (1 Jun) — 480 GB ceiling, 350 W, FP4→FP64 range	[I]
Customer sampling	H2 2026	[I]
Limited shipment	End of 2026 (Intel DC lead, via FT); broad availability 2027	[3P]
Cadence commitment	Annual data-centre GPU releases going forward	[I]
Sibling silicon	Xe3P also targeted at client Arc C-Series and Nova Lake iGPU	[I]

THE STRATEGIC BET IN ONE SENTENCE

By using LPDDR5X instead of HBM, Intel gives up roughly 3–5× peak memory bandwidth but gains 1.7–3.3× capacity per card, sidesteps the HBM allocation queue entirely, consumes no advanced-packaging interposer capacity, and lands inside a 350 W air-cooled envelope that drops into existing 4U/5U chassis without a facilities project. For inference workloads that are *capacity*-bound rather than *bandwidth*-bound — long context, large KV working sets, sparse MoE, many concurrent small agents — that is a favourable trade. For dense low-latency decode it is not.

2 • Confirmed specifications

PARAMETER	REFERENCE DESIGN	PARTNER CEILING	SRC
GPU architecture	Xe3P	—	[I]
Memory type	LPDDR5X (no HBM, no GDDR)	LPDDR5X	[I]
Memory capacity	160 GB	480 GB	[I]
Memory bandwidth	"up to 1.5 TB/s" (OCP 2025 slide)	—	[L]
Board power target	350 W	350 W class	[I]
Cooling	Air	Air	[I]
Form factor	PCIe add-in card, full-height	—	[I]
Host interface	PCIe (Gen5 x16 assumed)	—	[D]
Power connector	Single 16-pin (12V-2x6)	Partner-defined	[L]
Memory sites on PCB	20 packages (12 front, 8 rear)	20	[L]
VRM phases	18 positions, ~13 populated	—	[L]
Data types	FP4/MXFP4 → FP64	—	[I]
Compute throughput	<i>Not disclosed</i>	<i>Not disclosed</i>	[I]
Die size / node	<i>Not disclosed</i> ; die visibly larger than BMG-G31	—	[L]
Scale-up fabric	<i>Not disclosed</i> ; no proprietary fabric announced	—	[I]
Price	<i>Not disclosed</i>	—	[I]

ANOMALY WORTH TRACKING: THE BANDWIDTH CLAIM

Two incompatible bandwidth figures circulate. Chips and Cheese cites an Intel OCP 2025 presentation stating **up to 1.5 TB/s**; a leaked 1280-bit / LPDDR5X-9600 configuration reproduces that (1,536 GB/s) almost exactly. Tom's Hardware independently assumed a 640-bit bus at 10.7 Gbps and arrived at **684 GB/s** — a 2.2× discrepancy. Both appear in reputable outlets. \$6 models both. Until Intel publishes, treat the bandwidth of this part as a *range*, because the two ends of that range imply entirely different workload placement decisions.

3 · Xe3P architecture and the Xe-core

Xe3P is not a clean-sheet design. It is the performance/scalability variant of the Xe3 core shipping in Panther Lake, which itself preserves the Xe2 (Battlemage) core layout. This lineage is what makes a credible throughput derivation possible at all: the per-core execution resources have been stable across three generations, and Intel publishes exact numbers for the Xe2 discrete parts.

LEVEL	XE2 (BATTLEMAGE)	XE3 (PANTHER LAKE)	XE3P (CRESCENT ISLAND)
Xe Vector Engines / Xe-core	8 × 512-bit	8 × 512-bit	8 (assumed)
XXM matrix engines / Xe-core	8 × 2048-bit	8 × 2048-bit	8 (assumed)
Ray-tracing unit / Xe-core	1	1	1 (irrelevant to this use case)
SIMD width	SIMD16 native	SIMD16 native, +25% threads	SIMD16 assumed
Register allocation	Fixed	Variable	Variable
Xe-cores per render slice	4	6	6 (assumed)
Shared L1/SLM	Baseline	+33% vs Xe2	≥ Xe3
L2 cache	18 MB (BMG-G31 class)	16 MB on 12-core iGPU	Not disclosed; expect large
FP64	XVE extended-math block	Hardware-native FP64 restored	FP64 confirmed by Intel
Largest known config	32 Xe-cores (BMG-G31)	12 Xe-cores (iGPU)	Not disclosed

3.1 Calibration anchor — Intel Arc Pro B70

The B70 is the largest Xe2 discrete part with fully published throughput, making it the correct calibration point. Two independent Intel figures pin the per-core-per-clock rates exactly:

QUANTITY	INTEL PUBLISHED	MODEL	MODEL RESULT	MATCH
Xe-cores	32	—	—	—
Max clock	2,800 MHz	—	—	—
FP32 (vector)	22.94 TFLOPS	32 × 2.8 GHz × 256 ops	22.94 TFL0PS	✓ exact
INT8 peak (XXM, dense)	367 TOPS	32 × 2.8 GHz × 4,096 ops	367 T0PS	✓ exact
Memory	32 GB GDDR6, 256-bit, 608 GB/s	—	—	—
TBP	230 W (160–290 W range)	—	—	—

Source: Intel ARK SKU 245797, Arc Pro B70 datasheet. Intel footnotes its TOPS figure as *peak XXM throughput, INT8, dense models* — i.e. no sparsity multiplier. All derived figures below inherit that same dense convention.

4 · Data type support matrix

Intel's public claim is a contiguous range from FP4/MXFP4 up to FP64 — unusually broad for an inference-first part, and the FP64 end is what distinguishes it from a pure tokens-per-dollar accelerator.

DATA TYPE	ENGINE	REL. RATE	OPS/XE-CORE/CLK	STATUS	PRIMARY USE
FP4 / MXFP4	XXM	8×	8,192	[I] named at Computex	Max-throughput inference; MoE experts; weight-only quant
INT4	XXM	8×	8,192	[D] Xe2 precedent	AWQ/GPTQ weight-only quantisation
FP8 (E4M3/E5M2)	XXM	4×	4,096	[D] Xe3 adds native FP8	Mainstream serving precision; KV cache
INT8	XXM	4×	4,096	[I] Xe2 published	Classic quantised CNN/LLM inference
BF16	XXM	2×	2,048	[D] Xe2 precedent	Reference-accuracy serving; fine-tune
FP16	XXM	2×	2,048	[D] Xe2 precedent	Attention accumulate; activations
TF32	XXM	1×	1,024	[D] Xe2 precedent	Drop-in FP32 acceleration
FP32	XVE vector	0.25×	256	[I] Xe2 published	Non-matrix ops; softmax; norms; embeddings
FP64	XVE extended	0.03×	32 (1:8 est)	[I] supported, rate unknown	HPC / scientific; likely 1:8 or 1:16 of FP32
INT2	XXM	8×+	≥8,192	[L] Xe2 supported it	Research; rarely used in production

WHY THE FP4/MXFP4 LINE MATTERS MOST

On a bandwidth-limited card, the precision decision is primarily a *memory* decision, not a compute decision. Moving a 70 B model from BF16 to MXFP4 cuts its resident footprint from 141 GB to 40 GB and cuts per-token weight traffic by the same 3.6×. Because Crescent Island's compute-to-bandwidth ratio is high relative to HBM parts (\$7), low-precision formats convert almost linearly into tokens/s here — more so than on an HBM accelerator where compute saturates first. MXFP4 support is therefore the single most load-bearing item in the data-type list.

OPEN QUESTION: IS FP8 NATIVE IN XXM?

Intel documented Xe3 as adding FP8 *dequantization* support in the vector engine. At least one architecture analysis (HWCooling) reads this as the XXM matrix units still lacking native FP8 *compute*, with FP8 weights dequantised to INT8/FP16 paths. Throughput would be unaffected if the INT8 path is used, but numerics and kernel coverage would differ. Intel's explicit Computex "FP4 to FP64" framing for Xe3P suggests this is resolved in the data-centre part, but it is unconfirmed. Verify against the launch ISA documentation before committing an FP8-native serving plan.

5 • Derived compute throughput

NOTHING IN THIS SECTION IS AN INTEL SPECIFICATION

Intel has disclosed neither Xe-core count nor clock. The table below is a transparent scenario model: it applies the Xe2-calibrated per-core rates from §3.1 to three plausible Xe-core/clock configurations. Replace the two input variables when Intel publishes and every downstream number updates in one step.

$$\text{Throughput(dtype)} = N_{\text{xe-cores}} \times f_{\text{clock}} \times \text{ops_per_core_per_clock(dtype)}$$

where ops_per_core_per_clock is calibrated exactly against Arc Pro B70 (§3.1) and assumes Xe3P preserves the 8-XVE + 8-XXM Xe-core layout of Xe2/Xe3.

5.1 Scenario definitions

Perf/W column assumes the full 350 W board budget scales against the B70's 230 W TBP. In practice 160–480 GB of LPDDR5X draws a meaningful share of the 350 W, so the silicon-only budget is tighter than shown — biasing toward scenarios A and B.

SCENARIO	XE-CORES	CLOCK (GHZ)	XXM ENGINES	COMPUTE VS B70	IMPLIED PERF/W VS B70
CI-A Conservative	64	2.0	512	1.43×	0.94×
CI-B Central	96	2.0	768	2.14×	1.41×
CI-C Aggressive	128	2.2	1024	3.14×	2.07×

5.2 Peak dense throughput by data type

All values in TFLOPS (TOPS for integer types), dense — no sparsity multiplier. Reference column is calibrated against Intel-published B70 figures.

DATA TYPE	OPS/CORE/CLK	CI-A 64C@2.0	CI-B 96C@2.0	CI-C 128C@2.2	ARC PRO B70 (REF)
FP64 (XVE, 1:8 est)	32	4.1	6.1	9.0	2.9
FP32 (XVE vector)	256	32.8	49.2	72.1	22.9
TF32 (XXM)	1,024	131	197	288	91.8
FP16 / BF16 (XXM)	2,048	262	393	577	184
FP8 / INT8 (XXM)	4,096	524	786	1,153	367
FP4 / MXFP4 / INT4 (XXM)	8,192	1,049	1,573	2,307	734

PLANNING RECOMMENDATION

Use **CI-B as the central case**: roughly **390 TFLOPS BF16 / 790 TFLOPS-equivalent FP8 / 1,570 TFLOPS MXFP4**, with a ±40% band spanning CI-A to CI-C. CI-B implies a ~1.4× perf/W improvement over Battlemage at 1.52× the board power, which is a reasonable expectation for an architecture generation plus a node move, and is consistent with a die described as "visibly larger than BMG-G31". Anything materially above CI-C would be inconsistent with a 350 W air-cooled envelope that must also feed 20 LPDDR5X packages.

6 • Memory subsystem and physical layout

The memory subsystem is the product. Everything else is in service of putting the largest possible working set behind a single PCIe slot at the lowest cost per gigabyte.

6.1 Physical topology

ELEMENT	DETAIL	SRC
Package count	20 LPDDR5X sites — 12 on the PCB front, 8 on the rear	[L]
Per-device width	LPDDR5X die = 2 × independent 16-bit channels = 32-bit; 2-die package = 64-bit	[3P]
Aggregate bus	20 × 64-bit = 1,280-bit (leak-aligned) — alternative reading is 640-bit	[L]
Butterfly mode	Not available on LPDDR5X (unlike GDDR6/7) — constrains routing topology	[3P]
Placement consequence	Slow interface tolerates long traces; memory can sit on the rear half of the board rather than crowding the die	[3P]
Packaging impact	No silicon interposer / CoWoS consumed — decoupled from HBM supply constraints	[3P]
ECC	Expected (Arc Pro B-series already ships ECC); unconfirmed for this part	[D]

6.2 Bandwidth scenarios

M1 reproduces Intel's OCP 2025 "up to 1.5 TB/s" claim exactly and matches the leaked 20-package layout. M4 is Tom's Hardware's published 684 GB/s — which does not reconcile with the 640-bit bus that article cites (that yields 856 GB/s, shown as M3); 684 implies a 512-bit interface. M4 is used as the downside case throughout.

SCENARIO	STATED BUS (BIT)	GBPS/PIN	GB/S	GB/S PER PKG	TB/S	IMPLIED BUS (BIT)
M1 Reference / leak-aligned	1280	9.6	1,536	76.8	1.54	1,280
M2 Mid	1280	8.533	1,365	68.3	1.37	1,280
M3 Narrow bus	640	10.7	856	42.8	0.86	640
M4 Published downside	640	10.7	684	34.2	0.68	511

6.3 Capacity ladder

"Sweeps/s" = memory bandwidth ÷ capacity: how many times per second the card can read its entire memory. This is the single most useful number for reasoning about dense-model decode ceilings.

CONFIGURATION	PER PACKAGE	PER CARD	8-CARD NODE	SWEEPS/S (M1)	SWEEPS/S (M4)
Reference	8 GB	160 GB	1,280 GB	9.6	4.3
Mid-density	16 GB	320 GB	2,560 GB	4.8	2.1
Max partner	24 GB	480 GB	3,840 GB	3.2	1.4

6.4 Bandwidth-to-capacity balance vs the field

Competitor figures are nominal published specs, included for ratio context only — not a performance comparison.

PART	CAPACITY (GB)	BANDWIDTH (GB/S)	SWEEPS/S	SRC
Crescent Island 160 GB	160	1,536	9.6	[D]
Crescent Island 320 GB	320	1,536	4.8	[D]
Crescent Island 480 GB	480	1,536	3.2	[D]
Crescent Island 480 GB (M4 downside)	480	684	1.4	[D]
Intel Arc Pro B70	32	608	19.0	[I]
NVIDIA H200 SXM	141	4,800	34.0	[3P]
AMD MI355X	288	8,000	27.8	[3P]

THE DEFINING CHARACTERISTIC

At 480 GB, Crescent Island reads its full memory roughly **3.2 times per second**, against ~34 for an H200 and ~28 for an MI355X — an order of magnitude fewer. This is the whole story of the part in one ratio. A dense model that fills the card will decode slowly no matter how much compute is behind it. A sparse MoE that touches 5% of its weights per token, or a workload dominated by prefill and KV residency rather than autoregressive decode, sees almost none of that penalty. **Workload placement on this card is a sparsity and batching decision, not a capacity decision.**

7 • Roofline and balance analysis

Combining the §5 compute band with the §6 bandwidth band gives the arithmetic-intensity ridge point — the operations per byte a kernel must sustain before it stops being memory-bound.

Ridge point = peak compute ÷ peak bandwidth, at M1 bandwidth (1,536 GB/s) unless stated. Batch-1 autoregressive decode has arithmetic intensity near 1–2 op/B; prefill and large-batch decode reach hundreds.

SCENARIO	BF16 TFLOPS	RIDGE BF16 (FLOP/B)	RIDGE FP8 (OP/B)	RIDGE MXFP4 (OP/B)	RIDGE FP8, M4 BW
CI-A Conservative	262	171	341	683	767
CI-B Central	393	256	512	1024	1150
CI-C Aggressive	577	375	751	1502	1686

The practical reading: at every scenario, batch-1 decode sits two to three orders of magnitude below the ridge — deeply memory-bound, as on every other inference accelerator. The difference is *where the ceiling lands*. Because Crescent Island's ridge is high (compute is comparatively plentiful, bandwidth comparatively scarce), the card rewards aggressive batching and sparsity far more than an HBM part does, and punishes low-concurrency dense serving far more. Continuous batching depth is the primary tuning lever.

Batch-1 decode ceiling (weight-bound):

tokens/s ≈ memory_bandwidth ÷ (active_weight_bytes + KV_bytes_touched)

Concurrency needed to reach the compute roof:

batch* ≈ ridge_point (op/B) – i.e. one op/B per unit of batch, approximately

8 • Model fit

8.1 Weight residency

Weight footprint only; MXFP4 assumed 0.56 B/param including microscaling metadata. Add KV cache (§8.3), activations and framework overhead — budget ~10% headroom.

MODEL	ARCH	TOTAL (B)	ACTIVE (B)	BF16 GB	FP8 GB	MXFP4 GB
Llama 3.1 8B	Dense	8	8.0	16	8	4
Qwen3 32B	Dense	33	32.8	66	33	18
Llama 3.3 70B	Dense	71	70.6	141	71	40
gpt-oss-120b	MoE	117	5.1	234	117	66
Qwen3 235B-A22B	MoE	235	22.0	470	235	132
Llama 3.1 405B	Dense	405	405.0	810	405	227
DeepSeek-V3 / R1 671B	MoE (MLA)	671	37.0	1,342	671	376
Kimi K2 1T-A32B	MoE (MLA)	1,000	32.0	2,000	1,000	560

8.2 What fits where

Cell shows the highest-fidelity precision that leaves ≥15% of card memory free for KV cache, activations and fragmentation. "—" means no supported precision fits. Multi-card columns assume tensor/pipeline parallel with no replication.

MODEL	1× 160 GB	1× 320 GB	1× 480 GB	4× 320 GB (1.28 TB)	8× 480 GB (3.84 TB)
Llama 3.1 8B	BF16	BF16	BF16	BF16	BF16
Qwen3 32B	BF16	BF16	BF16	BF16	BF16
Llama 3.3 70B	FP8	BF16	BF16	BF16	BF16
gpt-oss-120b	FP8	BF16	BF16	BF16	BF16
Qwen3 235B-A22B	MXFP4	FP8	FP8	BF16	BF16
Llama 3.1 405B	—	MXFP4	FP8	BF16	BF16
DeepSeek-V3 / R1 671B	—	—	MXFP4	FP8	BF16
Kimi K2 1T-A32B	—	—	—	FP8	BF16

8.3 KV cache budget

KV bytes/token
=
2 × n_layers × n_kv_heads × head_dim × bytes_per_element

Max context tokens
=
(card_capacity − weight_footprint − overhead) ÷ KV bytes/token

20 GB reserved for activations, graph buffers and fragmentation. Rows marked multi-card do not fit on a single 480 GB card at any supported precision. KV tokens is the aggregate pool across all concurrent sequences — divide by batch size for per-request context.

MODEL	KV GEOMETRY	KV KB/TOK BF16	KV KB/TOK FP8	WT. PREC	FREE GB ON 480 GB CARD	KV TOKENS
Llama 3.1 8B	32L × 8 KV-heads × 128d	128	64	FP8	452	7,406K
Qwen3 32B	64L × 8 KV-heads × 128d	256	128	FP8	427	3,500K
Llama 3.3 70B	80L × 8 KV-heads × 128d	320	160	FP8	389	2,552K
gpt-oss-120b	36L × 8 KV-heads × 64d	72	36	FP8	343	9,991K
Qwen3 235B-A22B	94L × 4 KV-heads × 128d	188	94	FP8	225	2,510K
Llama 3.1 405B	126L × 8 KV-heads × 128d	504	252	FP8	55	229K
DeepSeek-V3 / R1 671B	MLA compressed latent	69	34	MXFP4	84	2,575K
Kimi K2 1T-A32B	MLA compressed latent	69	34	multi-card	—	—

WHERE THIS CARD IS GENUINELY DIFFERENTIATED

A 480 GB card running Llama 3.3 70B at FP8 leaves roughly 390 GB for KV — at 160 KB/token that is roughly **2.5 million cached tokens**. That is enough to hold hundreds of concurrent long-context agent sessions resident with no eviction, no recompute and no offload to a C1 flash tier. For agentic workloads where the same conversation is revisited across many turns, eliminating prefill recompute is often a larger throughput win than raw decode rate — which is precisely the regime where Crescent Island's capacity advantage converts into real tokens/s despite its bandwidth deficit.

8.4 Decode throughput ceilings (batch 1, M1 bandwidth)

Single-stream upper bound = bandwidth ÷ active weight bytes per token; ignores KV traffic, kernel efficiency and attention cost, so real-world figures land roughly 40–70% of these. Aggregate node throughput scales with batching well beyond these per-stream numbers.

MODEL	ARCH	ACTIVE FP8 GB	TOK/S FP8	ACTIVE MXFP4 GB	TOK/S MXFP4	TOK/S MXFP4 (M4 BW)
Llama 3.1 8B	Dense	8.0	192	4.5	343	153
Qwen3 32B	Dense	32.8	47	18.4	84	37
Llama 3.3 70B	Dense	70.6	22	39.5	39	17
gpt-oss-120b	MoE	5.1	301	2.9	538	239
Qwen3 235B-A22B	MoE	22.0	70	12.3	125	56
Llama 3.1 405B	Dense	405.0	4	226.8	7	3
DeepSeek-V3 / R1 671B	MoE (MLA)	37.0	42	20.7	74	33
Kimi K2 1T-A32B	MoE (MLA)	32.0	48	17.9	86	38

The MoE rows are the argument for this architecture. `gpt-oss-120b` touches only ~5.1 B parameters per token; at MXFP4 that is under 3 GB of weight traffic, giving a single-stream ceiling in the hundreds of tokens/s from a card that would struggle to reach 20 tok/s on a dense 405 B model. Dense models above ~100 B are the wrong workload for this part unless sharded across enough cards that per-card active weights fall back into the tens of gigabytes.

9 • Software stack

Intel's stated position is that the stack is "open, upstreamed, and Day 0 ready", with the Arc Pro B-series serving as the deliberate proving ground so that Crescent Island inherits a matured software base rather than bootstrapping one. That strategy is sound and observable — but independent reviews of the B60/B70 through mid-2026 consistently describe hardware that is ahead of its stack. Treat software maturity as the primary adoption risk, not silicon.

LAYER	COMPONENT	ROLE	MATURITY SIGNAL
Kernel / driver	Linux <code>xe</code> DRM driver	Device enumeration, memory management, scheduling	Upstream in mainline; Xe3-class enablement ongoing
Runtime	Intel Compute Runtime (Level Zero, OpenCL)	Low-level device API, command submission	Mature on Battlemage; standard path
Programming model	oneAPI / SYCL (DPC++)	Cross-vendor C++ heterogeneous model, open spec	Mature; far smaller ecosystem than CUDA
Compiler	Intel Graphics Compiler (IGC) + <code>icpx</code>	SPIR-V / GPU codegen	Mature
Kernel DSL	<code>intel-xpu-backend-for-triton</code>	Triton kernels on XPU — key for vLLM/PyTorch parity	Active; the critical path for new attention kernels
Primitives	oneDNN, oneMKL	GEMM, conv, attention primitives	Mature
Collectives	oneCCL	Multi-GPU all-reduce / all-gather	Functional; PCIe P2P dependent
Framework	PyTorch <code>xpu</code> device	Native upstream backend (<code>torch.xpu</code>)	Upstream; no out-of-tree fork required
Framework	IPEX (Intel Extension for PyTorch)	Ahead-of-upstream optimisations	Being folded into upstream over time
Graph runtime	OpenVINO	Optimised inference runtime, NNCF quantisation	Most mature Intel inference path
Graph runtime	ONNX Runtime (OpenVINO EP)	Portable ONNX execution	Mature
LLM serving	vLLM XPU backend	Paged attention, continuous batching, TP	Upstream collaboration active since 2025
LLM serving	<code>intel/llm-scaler</code>	Containerised vLLM distribution + multi-GPU	Shipping; <code>vllm-0.14.0-b8.2</code> added B70 support Apr 2026
Agentic	LangChain / Hugging Face / OPEA	Unmodified-code agentic orchestration	Intel's stated "just works" target
Orchestration	Kubernetes + Intel Device Plugins	Scheduling, allocation, node feature discovery	Available; standard container path
Manageability	ECC, SR-IOV, telemetry, remote FW update	Enterprise reliability and multi-tenancy	Delivered in Battlematrix; expected to carry over

9.1 Battlematrix / LLM-Scaler — the proving ground

Project Battlematrix is the multi-GPU Arc Pro programme whose software output is the direct ancestor of the Crescent Island stack. Its track record is the best available proxy for what launch-day software will look like.

MILESTONE	DATE	CONTENT
LLM Scaler 1.0	Aug 2025	First stable container; up to 1.8× TPOT on 32 B at >4K sequences, 4.2× on 70 B at 40K; PCIe P2P; ECC, SR-IOV, telemetry, remote firmware
vLLM Intel Arc Pro enablement	Nov 2025	Upstream vLLM collaboration: MoE optimisations, persistent kernels, multi-GPU scaling, LoRA, speculative decoding, structured output
StorageReview B60 8-GPU preview	Dec 2025	Setup described as straightforward; but only MXFP4 (gpt-oss) worked — INT4, FP8 and AWQ failed to start; only tensor parallelism functional
<code>vllm-0.14.0-b8.2</code>	Apr 2026	Official Arc Pro B70 support; image <code>intel/llm-scaler-platform:26.18.8.2</code>
StorageReview B70 review	~Jul 2026	"The hardware is ready, the stack needs to catch up" — the recurring verdict

THE SINGLE BIGGEST ADOPTION RISK

In December 2025 testing of an 8-GPU Battlematrix system, only MXFP4 models loaded successfully; INT4, FP8 and AWQ quantisations failed outright, and expert parallelism and pipeline parallelism were non-functional — only tensor parallelism worked. Quantisation-format coverage and parallelism-strategy coverage are exactly the capabilities that determine whether a 480 GB card can be used at its capacity. This has improved through 2026, but it should be an explicit acceptance-test gate in any evaluation: **bring your own model, in your own quantisation, with your intended parallelism strategy**, and validate on sampling silicon before committing volume.

9.2 Recommended evaluation checklist

#	GATE	PASS CRITERION
1	Model loads in target quantisation	MXFP4, FP8, INT4/AWQ, GPTQ all load without fallback
2	Parallelism strategies	TP confirmed; EP and PP required for MoE at scale
3	Multi-card scaling	PCIe P2P enabled; measure all-reduce bandwidth, not just card count
4	Long-context behaviour	TTFT and TPOT at 32K/128K, not just short prompts
5	Continuous batching depth	Throughput curve to batch 64+ — this card needs depth (\$7)
6	KV cache offload / prefix reuse	Validate prefix caching works; it is the main agentic win
7	Telemetry	Per-card utilisation, memory, power, thermals exposed to the scheduler
8	Multi-tenancy	SR-IOV partitioning and ECC reporting under load
9	Framework parity	Target model's attention kernel present in the Triton XPU backend
10	Air-cooling headroom	Sustained 350 W in the intended 4U/5U chassis at inlet temp

10 · Platform integration with Xeon 6

Crescent Island is a PCIe endpoint with no announced proprietary scale-up fabric, which places the entire multi-card burden on the host's PCIe topology. That makes host CPU selection consequential in a way it is not for NVLink-domain accelerators: on this platform, the CPU root complex *is* the scale-up fabric.

Node Element	Requirement	Rationale
Host CPU	Xeon 6 6700P/6900P class with sufficient PCIe Gen5 lanes	8 cards × x16 = 128 lanes of GPU I/O before NIC and storage
PCIe generation	Gen5 x16 per card, P2P enabled in BIOS	P2P is mandatory for tensor-parallel all-reduce performance
Host memory	≥2× aggregate GPU memory for staging and model loading	480 GB cards imply very large model artefacts to page in
Host memory bw	MRDIMM where available	Sustained model load and KV spill/fill traffic
Chassis	4U/5U air-cooled, 8 cards × 350 W = 2.8 kW GPU power	Plus CPU, NIC and fans — verify rack power and inlet temperature
Storage (C0/C1)	NVMe capable of saturating model-load and KV-spill paths	Cross-reference Storage Architecture Reference, C1 KV tier
Network	Scale-out KV transfer plane sized to disaggregated prefill/decode	Cross-reference Networking Reference, plane 2

10.1 Host CPU candidates from the SKU workbook

The workbook's [AI / GenAI Infrastructure → Inference of Large GenAI models \(GPU host CPU\)](#) rows are the correct starting point. Crescent Island shifts the emphasis relative to an HBM accelerator node: with no proprietary fabric, PCIe lane count and root-complex behaviour matter more than raw core count, and the eight-card configuration is I/O-dominant.

Row numbers reference [Intel_Xeon6_Workload_SKU_Recommendations.xlsx](#). For vector-search co-residency alongside inference, see workbook rows 83–86 (Vector Search & Indexing), where the 6787P/86-core option applies.

Tier	SKU	Cores	TDP (W)	Workbook Row	Fit for a Crescent Island Node
Good	6732P	32	350	78	Adequate for 4-card nodes; lane budget is the constraint
Good	6747P	48	330	77	Balanced entry point for 4–8 card air-cooled builds
Better	6767P	64	350	80	Recommended baseline for 8-card nodes
Better	6960P	72	500	79	Where host-side pre/post-processing is heavy
Best	6776P*	64	350	82	Best perf/W at 8 cards; preferred if power-constrained
Best	6962P*	72	500	81	Maximum host headroom; verify rack power budget

10.2 Node-level aggregate (8 cards)

Aggregate bandwidth is the sum of independent per-card figures — it is not achievable across a tensor-parallel model, where PCIe interconnect becomes the limiter. Compute figures use the CI-B central scenario and are derived, not published.

Node Config	GPU Memory	Aggregate BW (M1)	FP8 TFL0PS (CI-B)	MXFP4 TFL0PS (CI-B)	GPU Power
8 × 160 GB	1,280 GB	12.3 TB/s	6,291	12,583	2,800 W
8 × 320 GB	2,560 GB	12.3 TB/s	6,291	12,583	2,800 W
8 × 480 GB	3,840 GB	12.3 TB/s	6,291	12,583	2,800 W

11 • Workload placement guidance

WORKLOAD	FIT	REASONING
Sparse MoE serving (gpt-oss, Qwen3-A22B, DeepSeek)	Strong	Low active-parameter fraction hides the bandwidth deficit; capacity holds the full expert set resident
Long-context agentic sessions with prefix reuse	Strong	Massive KV pool eliminates eviction and prefill recompute — the dominant agentic cost
High-concurrency batch inference / offline generation	Strong	Throughput-oriented; batching pushes toward the compute roof where the card is well provisioned
Many small models co-resident (agent swarms)	Strong	3.84 TB per node hosts dozens of specialised models without cross-node routing
Embedding and reranker serving	Strong	Prefill-dominant, compute-bound, minimal decode — plays to the card's strengths
RAG pipelines with in-memory vector index	Good	Large capacity can host index shards; cross-check against DiskANN/AiSAQ cost model first
Dense 7B–70B serving at moderate concurrency	Good	Fits comfortably; decode rate is workmanlike, not leading — validate TPOT against SLA
Vision-language models	Good	Image prefill is compute-heavy and batches well; verify VLM kernel coverage in the stack
Speech / ASR / TTS	Good	Small models, high concurrency, capacity-tolerant
Dense 200B+ low-latency decode	Weak	Bandwidth-bound; sweeps/s ceiling caps per-stream tokens/s regardless of compute
Latency-critical single-stream chat (tight TTFT+TPOT SLA)	Weak	Batch-1 decode is the card's least favourable regime
Large-scale model training	Not targeted	No proprietary scale-up fabric; Intel does not position this part for training
FP64-heavy HPC	Conditional	FP64 is supported but the rate is undisclosed — likely 1:8 or 1:16; do not assume vector FP64 parity

12 · Risks and open questions

#	OPEN ITEM	WHY IT MATTERS	RESOLVE BY
1	Compute throughput undisclosed	Every TFLOPS figure in circulation is derived; \$5 has a $\pm 40\%$ band	Intel launch disclosure / SC26
2	Bandwidth: 684 GB/s or 1.54 TB/s (sources disagree, and one is self-inconsistent)	2.2× discrepancy inverts several workload-placement decisions	Launch specification
3	Native FP8 in XMV vs dequant-only	Determines numerics and kernel coverage for the mainstream serving precision	Launch ISA documentation
4	FP64 rate (1:8, 1:16, or scalar-only)	Decides whether any HPC claim is credible	Launch specification
5	Quantisation-format coverage in the stack	Dec 2025 testing showed only MXFP4 loading on Battlematrix	Hands-on evaluation on sampling silicon
6	Expert and pipeline parallelism support	MoE at 480 GB scale needs EP; only TP was functional in late 2025	Hands-on evaluation
7	No announced scale-up fabric	PCIe P2P becomes the all-reduce path; caps tensor-parallel efficiency	Launch platform documentation
8	LPDDR5X supply	DRAM shortage conditions through 2026; 20 packages/card at 24 GB is a large bill	Supply commitment at contract time
9	480 GB availability at launch	Intel frames 480 GB as a partner ceiling; 160/240 GB likely ship first	Partner SKU announcements
10	ECC confirmation	Assumed from Arc Pro precedent but unconfirmed for this part	Launch specification
11	Pricing	The entire thesis is cost per GB and per token — undisclosed	Launch
12	Die size, node, fab	Affects supply, cost and the plausibility of the CI-C scenario	Launch / die analysis

13 · Assumptions, methodology and sources

13.1 Explicit assumptions

#	ASSUMPTION	BASIS	SENSITIVITY
A1	Xe3P preserves the 8 XVE + 8 XMX Xe-core layout	Stable across Xe2 and Xe3	High — invalidates all of \$5 if wrong
A2	XMX per-core-per-clock rates match Xe2	Exact calibration to B70 published FP32 and INT8	High
A3	MXFP4/FP4 runs at 2× the INT8 rate	Standard microscaling implementation ratio	Medium — affects \$5 bottom row only
A4	FP64 at 1:8 of FP32	Typical XVE extended-math ratio; could be 1:16	Low — HPC only
A5	1,280-bit bus at LPDDR5X-9600	Reproduces Intel's "up to 1.5 TB/s" exactly; matches 20-package leak	High
A6	Peak theoretical, dense, at stated clock	Same convention as Intel's B70 TOPS footnote	Medium — real efficiency 40–70%
A7	MXFP4 weights at 0.56 B/param	4-bit plus microscaling metadata	Low
A8	20 GB reserved for non-weight, non-KV state	Framework overhead, activations, fragmentation	Low
A9	PCIe Gen5 x16 host interface	Consistent with 16-pin power and Arc Pro B70 precedent	Medium
A10	ECC present	Arc Pro B-series precedent	Low

13.2 What Intel has actually published vs what is derived

CATEGORY	PUBLISHED BY INTEL	DERIVED / LEAKED HERE
Architecture	Xe3P; performance-per-watt focus; agentic AI positioning	Xe-core layout, core count, clock
Memory	LPDDR5X; 160 GB reference; 480 GB partner ceiling	Bus width, per-pin rate, exact bandwidth, package layout
Power / thermal	350 W; air-cooled; PCIe AIC	Cooling headroom, node aggregate
Data types	FP4/MXFP4 through FP64	Per-type throughput ratios and absolute rates
Compute	Nothing	All of \$5
Software	oneAPI; "open, upstreamed, Day 0 ready"; Arc Pro B-series proving ground	Layer-by-layer maturity assessment
Timeline	H2 2026 sampling; annual cadence	2027 broad availability
Price	Nothing	—

13.3 Sources

#	SOURCE	USED FOR	CLASS
S1	Intel Newsroom — "Intel to Expand AI Accelerator Portfolio with New GPU", 14 Oct 2025	Announcement, positioning, sampling window, software strategy	[I]
S2	Intel Computex 2026 disclosures, 1 Jun 2026	480 GB ceiling, 350 W, FP4-FP64 range, agentic framing	[I]
S3	Intel ARK — Arc Pro B70 (SKU 245797) and B70 datasheet	Calibration anchor: 32 Xe-cores, 2,800 MHz, 22.94 TFLOPS FP32, 367 INT8 TOPS	[I]
S4	Intel Xe3 architecture disclosure (Panther Lake)	Xe-core composition, XMX width, FP8 dequant, FP64 restoration	[I]
S5	Tom's Hardware — Computex Crescent Island coverage, 1 Jun 2026	350 W, PCIe AIC, no throughput specs, 640-bit / 684 GB/s alternative	[3P]
S6	Tom's Hardware / VideoCardz / TechPowerUp — PCB leak coverage, May 2026	20 memory sites (12F/8R), 16-pin power, 18 VRM positions, die vs BMG-G31	[L]
S7	VideoCardz — LPDDR5X-9600 / 1.5 TB/s report, 21 May 2026	1,280-bit bus, 9,600 MT/s, ~1.54 TB/s	[L]
S8	Chips and Cheese, citing Intel OCP 2025 presentation	"Up to 1.5 TB/s" bandwidth claim	[3P]
S9	Phoronix — Crescent Island announcement; LLM-Scaler release notes	Xe3P lineage; vllm-0.14.0-b8.2 with B70 support, Apr 2026	[3P]
S10	StorageReview — Arc Pro B60 Battlematrix preview (Dec 2025) and B70 review	Quantisation-format failures, parallelism limits, stack maturity verdict	[3P]
S11	vLLM blog — Intel Arc Pro B-series enablement, Nov 2025	Upstream vLLM feature set: MoE, persistent kernels, LoRA, spec decoding	[3P]
S12	TechPowerUp / VideoCardz — Project Battlematrix LLM Scaler 1.0, Aug 2025	ECC, SR-IOV, telemetry, remote FW; 1.8×/4.2× long-sequence gains	[3P]
S13	HWCooling — Xe3 architecture analysis	FP8 XMX-native ambiguity	[3P]
S14	TechSpot / DataCenterDynamics / HPCwire — Computex and OCP coverage	2027 availability; no benchmarks disclosed	[3P]
S15	Financial Times, via secondary reporting	Limited shipment by end of 2026; 18-month development	[3P]
S16	Intel_Xeon6_Workload_SKU_Recommendations.xlsx	Host CPU SKU cross-references, rows 71–86	Internal

REVISION GUIDANCE

This document is built to be corrected cheaply. When Intel publishes Xe-core count and clock, update the two variables in §5.1 and the entire throughput chain (§5.2, §7, §10.2) follows. When bandwidth is confirmed, retire the losing scenario in §6.2 and §8.4 collapses to a single column. Everything in §8 and §11 is a function of those two quantities. Sections 9 and 12 should be re-checked against hands-on results from sampling silicon rather than from published material.

Compiled 9 August 2026. All third-party figures are as reported by the cited outlets at the dates shown; pre-launch specifications are subject to change without notice. Derived values are the author's modelling and carry no vendor endorsement.